Sri Sivasubramaniya Nadar College of Engineering, Chennai

(An Autonomous Institution affiliated to Anna University)

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Experiment 7

Bagging, Boosting, and Stacked Ensemble Models

Objective

- To understand ensemble learning strategies: Bagging, Boosting, and Stacking.
- To implement Bagging and Boosting classifiers.
- To build a Stacked Ensemble model using multiple base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization.
- To analyze the effect of ensemble methods on bias and variance.

Dataset

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant (M) and Benign (B)

Dataset Link: https://archive.ics.uci.edu

Theory

Bagging (Bootstrap Aggregation)

Bagging is an ensemble technique that trains multiple models on different bootstrap samples of the training data and aggregates their predictions.

Key Characteristics:

- Reduces variance
- Suitable for high-variance models
- Independent model training

Boosting

Boosting is an ensemble method where models are trained sequentially, and each new model focuses on correcting the errors made by previous models.

Key Characteristics:

- Reduces bias
- Sequential learning
- Emphasizes difficult samples

Common boosting algorithms include AdaBoost and Gradient Boosting.

Stacked Ensemble (Stacking)

Stacking combines multiple heterogeneous base models and trains a meta-learner to optimally combine their predictions.

Key Characteristics:

- Uses diverse base learners
- Meta-model learns optimal combination
- Often achieves superior performance

Steps for Implementation

1. Load and preprocess the dataset.
2. Perform Exploratory Data Analysis (EDA).
3. Split the dataset into training and testing sets (80–20).
4. Implement a Bagging classifier using a base estimator.
5. Implement Boosting classifiers (AdaBoost and/or Gradient Boosting).
6. Construct a Stacked Ensemble using multiple base models.
7. Define hyperparameter search spaces for each ensemble model.
8. Select hyperparameters using 5-fold cross-validation.
9. Evaluate all models using standard performance metrics.
10. Compare results and analyze bias–variance behavior.

Ensemble Models Used

Bagging

- Base Estimator: Decision Tree
- Number of estimators
- Sampling strategy

Boosting

- AdaBoost
- Gradient Boosting

Stacked Ensemble

- Base Models: SVM, Naïve Bayes, Decision Tree
- Meta Learner: Logistic Regression

Hyperparameters to be Explored

Bagging

- `n_estimators`
- `max_samples`
- `max_features`

Boosting

- `n_estimators`
- `learning_rate`
- `max_depth`

Stacked Ensemble

- Choice of base models
- Final estimator

Hyperparameter Evaluation Results

Bagging Results

Table 1: Bagging Hyperparameter Evaluation

n_estimators	max_samples	Avg CV Accuracy (%)	Avg CV F1 Score

Table 2: Boosting Hyperparameter Evaluation

n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score

Table 3: Stacked Ensemble Evaluation

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score

Boosting Results

Stacked Ensemble Results

Performance Comparison

Table 4: Performance Comparison of Ensemble Models

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging				
Boosting				
Stacked Ensemble				

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC Curve and AUC

Observation Questions

- How does Bagging reduce variance?
- How does Boosting address model bias?
- Why does stacking benefit from heterogeneous models?
- Which ensemble method performed best and why?

Conclusion

Bagging, Boosting, and Stacked Ensemble models were implemented and evaluated. The experiment demonstrates how different ensemble strategies improve predictive performance and generalization compared to individual models.

References

- Scikit-learn: Ensemble Methods
- Bagging Classifier
- UCI Dataset